**1.Project Identification cover Page**

**Advanced Grid-Integrated Solar PV System Design and Impact Analysis for Commercial and Industrial Facilities in Alberta**

**Team Number- Group 5**

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## 3. Glossary

The following terms, abbreviations, and nomenclature are used throughout this report:

| Term / Abbreviation | Definition |
|---|---|
| AC | Alternating Current - the standard form of electrical power delivered to the grid |
| BESS | Battery Energy Storage System - a system that stores electrical energy for later use |
| CAPEX | Capital Expenditure - upfront investment costs for purchasing and installing equipment |
| DC | Direct Current - the type of electricity generated by photovoltaic solar panels |
| DC/AC Ratio | The ratio of total DC capacity of PV modules to the rated AC output of the inverter |
| DG | Distributed Generation - power generated at or near the point of consumption |
| ETAP | Electrical Transient Analyzer Program – software used for power system modeling and analysis |
| IRR | Internal Rate of Return – the discount rate at which the net present value of a project equals zero |
| LCOE | Levelized Cost of Energy - average cost per kWh of electricity produced over the system lifetime |
| MPPT | Maximum Power Point Tracking - an algorithm used in inverters to maximize power extraction |

| Term / Abbreviation | Definition |
|---|---|
| MSB | Main Switchboard - the primary electrical distribution board in the TFDL |
| N-1 Contingency | A reliability criterion where the system must continue operating after losing any single component |
| NPV | Net Present Value - the present value of future cash flows minus the initial investment |
| O&M | Operations and Maintenance - ongoing costs associated with running and maintaining the system |
| OpenSolar | Cloud-based solar design and proposal software used for PV layout and energy modeling |
| PCC | Point of Common Coupling - the point where the PV system connects to the utility grid |
| PID | Potential Induced Degradation - performance loss in PV modules caused by voltage leakage |
| PV | Photovoltaic - technology that converts solar radiation into direct current electricity |
| p.u. | Per Unit - a normalized expression of electrical quantities relative to a defined base value |
| SLD | Single-Line Diagram - a simplified electrical diagram showing the main components of a power system |

| Term / Abbreviation | Definition |
|---|---|
| TFDL | Taylor Family Digital Library – the University of Calgary building selected as the case study site |
| TOPCon | Tunnel Oxide Passivated Contact – an advanced solar cell technology used in the selected PV modules |
| TX1 / TX2 | Transformer 1 / Transformer 2 – the two main transformers in the TFDL electrical system |
| Volt-VAR | A smart inverter control function that adjusts reactive power output based on voltage |
| Volt-Watt | A smart inverter control function that curtails active power output when voltage is too high |

**4. Executive Summary**

This report presents the complete design, analysis, and validation of an advanced grid-integrated solar photovoltaic (PV) system for the Taylor Family Digital Library (TFDL) at the University of Calgary, Alberta. The project was undertaken as part of the ENEL 678 Graduate Capstone Project in Electrical Engineering by Group 5.

The primary objective was to design a technically feasible, grid-compliant, and economically viable rooftop PV system that would meaningfully reduce the TFDL's dependence on utility grid power, lower carbon emissions, and provide a replicable framework for similar commercial and industrial facilities across Alberta.

Using OpenSolar, the team designed a 363.55 kW DC / 384 kW AC rooftop PV system comprising 661 Axitec 550W bifacial modules and seven Solis and Growatt grid-tied inverters. The system is projected to generate approximately 487,283 kWh of electricity annually, offsetting between 27% (winter) and 37% (summer) of the building's total utility demand. A detailed AutoCAD Single-Line Diagram (SLD) was

developed to illustrate the electrical integration of the PV system with TFDL's existing 600V Main Switchboard (MSB).

The electrical grid impact was rigorously assessed using ETAP power system simulation software. Load flow analysis, time-series simulations, contingency (N-1) studies, reverse power flow assessment, and smart inverter function evaluations were conducted. Results confirmed that the 384 kW AC system operates within acceptable voltage limits (0.99 p.u.), reduces transformer loading by 40–47%, and maintains grid stability under all modeled conditions. Volt-VAR and Volt-Watt smart inverter control functions were implemented and validated in ETAP, confirming effective voltage regulation.

An economic analysis determined a Levelized Cost of Energy (LCOE) of $0.079/kWh, compared to Alberta's commercial electricity rate of $0.12/kWh, confirming strong financial viability. The project yields an estimated Internal Rate of Return (IRR) of approximately 4.25% over a 25-year system lifetime. In addition, the system is estimated to reduce carbon emissions by approximately 228 metric tonnes of $CO_2e$ per year, based on Alberta's grid emission factor of 0.470 kg $CO_2e$/kWh.

A physical low-voltage prototype was developed using an Arduino Nano ESP32, ACS712 current sensor, and voltage divider circuit to conceptually validate real-time voltage monitoring, current measurement, and simulated inverter response. The prototype demonstrated the operational principles of the full-scale system.

All primary success criteria were met or exceeded, including PV capacity, annual energy generation, bus voltage stability, transformer loading reduction, LCOE, $CO_2$ reduction, and N-1 contingency stability. The project provides a strong foundation for future work, including battery energy storage system (BESS) integration, advanced protection coordination, and further financial modeling.

## 5. Project Motivation and Objectives

### 5.1 Project Motivation

Alberta's electricity sector is heavily reliant on fossil fuels, particularly natural gas, contributing significantly to greenhouse gas emissions. The provincial grid emission factor of 0.470 kg $CO_2e$/kWh underscores the environmental cost of grid-sourced electricity. Simultaneously, commercial and institutional buildings such as universities are among the largest electricity consumers, representing a significant opportunity for renewable energy integration.

The University of Calgary has committed to sustainability goals, including reducing its carbon footprint. The Taylor Family Digital Library (TFDL) was selected as the case study site due to its large, unobstructed

rooftop area, well-documented electrical infrastructure, and significant energy consumption. Despite this potential, the TFDL currently relies entirely on grid power, with no on-site renewable generation.

Beyond the environmental imperative, rising electricity costs and the maturing economics of solar PV technology make rooftop solar increasingly attractive for commercial facilities. Grid-connected PV systems with smart inverter controls can provide multiple benefits: reduced energy bills, improved grid reliability, lower carbon emissions, and long-term energy cost predictability.

However, integrating large-scale distributed generation into an existing commercial electrical system presents technical challenges, including voltage rise, reverse power flow, transformer overloading, and protection coordination issues. These challenges motivated the need for a rigorous, simulation-based engineering approach, using industry-standard tools such as ETAP and OpenSolar.

**5.2 Project Objectives**

The project was guided by the following specific, measurable, and time-bound objectives:

- Design a rooftop PV system with a capacity between 300 kW and 600 kW for the TFDL, optimized within the building's available rooftop area and electrical infrastructure constraints.
- Estimate annual energy generation using OpenSolar simulation tools and validate against building load data.
- Develop a detailed AutoCAD Single-Line Diagram showing the complete electrical integration of the PV system with the existing 600V MSB.
- Build and validate an ETAP power system model to perform load flow analysis, time-series simulation, reverse power flow assessment, N-1 contingency analysis, and smart inverter evaluation.
- Evaluate Volt-VAR and Volt-Watt smart inverter control functions to ensure voltage compliance within 0.95–1.05 p.u. at all grid connection points.
- Conduct an economic feasibility analysis, including LCOE, NPV, and IRR calculations, to assess the financial viability of the proposed system.
- Quantify the environmental impact of the system in terms of annual $CO_2$ equivalent reductions.
- Develop a low-voltage physical prototype to conceptually validate key operational principles, including real-time voltage and current monitoring, power calculation, and simulated inverter response.

## 6. Methodology and Design

### 6.1 Site Selection and Feasibility Assessment

The Taylor Family Digital Library (TFDL) at the University of Calgary (2500 University Drive NW, Calgary, Alberta) was selected based on its large south-facing rooftop area, well-documented electrical infrastructure, and substantial annual energy consumption. A site feasibility study confirmed the building's 2000 kVA transformer (13.2 kV / 347–600 V) and 600V Main Switchboard (MSB) could accommodate the proposed PV interconnection. Building load data, including hourly demand profiles, was obtained through direct communication with TFDL facility management.

### 6.2 PV System Design using OpenSolar

The rooftop PV system was designed using OpenSolar, a cloud-based platform for solar system modeling, layout design, and energy performance estimation.

The design process included the following:

- Module Selection: 661 Axitec AXIbiperfect GQ TS 550W bifacial N-Type modules were selected, resulting in a total DC capacity of approximately 363.55 kW. These modules offer high efficiency (22.33%), low degradation, and a 30-year warranty, with bifacial capability improving overall energy yield.
- Panel Placement: Modules were arranged in portrait orientation across the rooftop, considering shading, structural constraints, and layout optimization. The system was configured with an azimuth of ~158° and a tilt angle of 13°, suitable for Calgary conditions to maximize solar exposure.
- String Sizing and MPPT Configuration: Strings of 15–19 modules were configured per MPPT to maintain voltage within inverter operating limits under varying conditions. The DC/AC ratio was maintained within an acceptable range (0.82–1.05) to ensure efficient inverter utilization.
- Inverter Selection: The system includes five 60 kW Solis inverters, one 40 kW Solis inverter, one 30 kW Growatt inverter, and one 14 kW inverter, resulting in a total AC capacity of approximately 384 kW. This distributed configuration improves reliability and performance under partial shading.
- Energy Estimation: OpenSolar estimated an annual energy generation of approximately 487,283 kWh using Calgary-specific irradiance data, inverter efficiency, and system losses (15%, including 1.8% inverter loss and 1.1% shading loss).
- Model Validation and Adjustments: Certain outputs from OpenSolar, such as consumption offset and financial savings, were based on default assumptions and did not reflect actual building conditions. These values were validated and corrected using ETAP simulations and verified load data. The actual energy

offset was determined to be approximately 27–37%, and economic and environmental analyses were recalculated using Alberta-specific data.

Overall, OpenSolar provided the initial system design and energy estimation, which was further refined and validated through detailed ETAP analysis to ensure technical accuracy.

### 6.3 Electrical Integration and AutoCAD Single-Line Diagram

A detailed AutoCAD Single-Line Diagram (SLD) was developed to document the complete electrical integration of the PV system with the TFDL's existing 600V MSB. Key design elements included:

- Eight inverter feeder circuits connecting to a dedicated PV Distribution Board (PV Dist. Board) on the rooftop.
- DC isolators installed on the DC side of each inverter for safety and maintenance isolation.
- A generation kWh meter to record PV energy output.
- A lockable 2-pole AC isolator between the PV Distribution Board and the MSB.
- A dedicated main backfeed breaker at the 600V MSB rated at 21.18A for PV interconnection.
- AC cables routed from the rooftop PV Distribution Board down to the basement MSB, connecting through the existing TX2 2000 kVA transformer (13.2kV / 347-600V).

The SLD was developed in compliance with Canadian Electrical Code (CEC) requirements and standard industry practice for grid-connected PV systems.

### 6.4 ETAP Power System Modeling and Grid Impact Analysis

ETAP (Electrical Transient Analyzer Program) was used to build a detailed power system model of the TFDL electrical distribution system with the integrated PV system. The ETAP modeling and analysis process included:

#### 6.4.1 Model Development

The ETAP model was developed based on the AutoCAD SLD and the original TFDL record drawing (E600). To comply with ETAP's 25-bus limit for load flow analysis, multiple distributed loads were aggregated into equivalent lumped loads per floor while preserving the main network structure and transformer characteristics. The model included:

- Utility grid connection at 13.2 kV
- Two 2000 kVA step-down transformers (TX1 and TX2, 13.2 kV / 600 V)
- Distribution cables and bus connections representing the TFDL's six-floor electrical distribution

- Aggregated load buses representing building electrical demand
- Two PV generators (Inv1 and Inv2) representing the full 363 kW DC / 384 kW AC system

### 6.4.2 Load Flow Analysis

Load flow studies were conducted for three operating conditions Average, Peak, and Minimum load both with and without the PV system active. PV penetration was varied at 0%, 25%, 50%, 75%, and 100% to evaluate the impact on utility import, transformer loading, and bus voltages. Key findings:

| PV Penetration | Utility Power (kW) | TX1 Loading (%) | TX2 Loading (%) | System Behavior |
|---|---|---|---|---|
| 0% (No PV) | 463.1 | 87.28% | 85.92% | Full grid import |
| 25% | 368.8 | 80.37% | 78.34% | Reduced import |
| 50% | 312.3 | 66.56% | 66.02% | Moderate reduction |
| 75% | 261.4 | 46.05% | 44.53% | Significant reduction |
| 100% | 238.5 | 31.59% | 11.35% | Maximum reduction |
| Reduced Load + 100% | 185 (export) | Low | Low | Partial reverse power flow |

### 6.4.3 Time-Series Load Flow Analysis

Time-series simulations were performed for January 8 (winter, low irradiance) and June 8 (summer, high irradiance) to evaluate time-varying grid interactions under realistic load and PV generation conditions. Key findings:

- Peak utility demand was reduced from 608 kVA (base case) to 446 kVA in winter (27% reduction) and 380 kVA in summer (37% reduction).
- Maximum PV impact was observed between 10:30 AM and 1:30 PM, corresponding to peak solar irradiance hours.

- At 6:00 AM and 6:30 PM, PV contribution was negligible, and utility remained the primary power source.
- Summer months showed greater demand reduction due to higher available solar irradiance.

#### 6.4.4 N-1 Contingency Analysis

Two N-1 contingency scenarios were simulated — average load with TX1 out of service and minimum load with TX1 out of service combined with full PV generation:

- Case 1 (Average Load + TX1 Outage): TX2 successfully supplied both transformer buses via the closed tie switch. TX2 loading increased to 15–17% above normal but remained well within safe operating limits. Bus voltages remained within 0.992–0.995 p.u. No overload conditions observed.
- Case 2 (Minimum Load + TX1 Outage + Full PV): This represented the most critical operating condition. PV generation (343 kW) exceeded total building demand, causing reverse power flow from PV buses through TX2 to the utility. Bus voltages approached 0.998–1.00 p.u. No voltage violations were observed, but reverse power flow was confirmed.

#### 6.4.5 Smart Inverter Function Evaluation

Volt-VAR and Volt-Watt control functions were implemented in the ETAP inverter models using user-defined Q(V) and P(V) characteristic curves:

- Volt-VAR Control: The Q(V) curve was configured to inject reactive power (Q) when terminal voltage dropped below 0.95 p.u. and absorb reactive power when voltage exceeded 1.05 p.u. The inverter reference voltage was set at 100% (1.0 p.u.), with cut-off limits at 88%–110%.
- Volt-Watt Control: Active power curtailment was triggered when the PCC voltage exceeded 1.03 p.u., preventing voltage from exceeding the 1.05 p.u. grid limit. Under all simulated conditions, PCC voltage remained stable at ~0.99 p.u., confirming no voltage violations occurred.

These functions collectively confirmed improved voltage regulation and mitigation of voltage rise under high PV penetration conditions.

### 6.5 Physical Prototype Development

The physical prototype was developed to provide a simplified, low-voltage demonstration of the fundamental operating principles of a grid-integrated PV system. While the primary focus of this project is large-scale system simulation using ETAP and OpenSolar, the prototype serves as a conceptual validation tool to bridge theoretical analysis with practical implementation.

The prototype was constructed using an Arduino Nano ESP32 microcontroller, a ZMPT101B voltage sensing module, and an ACS712 current sensor. A regulated DC power supply (Jesverty SPS-3010N) was used to emulate PV output under varying voltage conditions. A resistive load was connected in series with the current sensor to ensure measurable current flow, enabling real-time monitoring of electrical parameters.

The Arduino-based system was programmed to perform continuous data acquisition and processing. Voltage and current signals were sampled using the onboard 12-bit analog-to-digital converter (ADC), with multiple samples averaged to reduce measurement noise. A zero-offset calibration routine was implemented at system startup to eliminate sensor bias, particularly for the ACS712 current sensor. Additionally, a dead-band threshold was applied to suppress false current readings near zero.

The prototype computes instantaneous power using the relation $P = V \times I$, and outputs real-time measurements via the serial monitor. An LED indicator was integrated to visually represent active power flow conditions.

To simulate inverter-like behavior, the system was tested under three operating conditions: low voltage, normal voltage, and high voltage. These conditions represent simplified analogs of real grid-tied inverter response regions, where protective or control actions would typically occur.

It is important to note that the prototype operates at low voltage (≤30 V DC) and does not replicate full-scale grid conditions. Its purpose is strictly conceptual, demonstrating measurement, monitoring, and basic control logic consistent with the simulated large-scale system.

## 7. Product Scope and Final Product Functionality

### 7.1 Defined Scope

The scope of this project was defined in the Design Roadmap and included the following key deliverables:

- A rooftop PV system design using OpenSolar, including module selection, string sizing, MPPT allocation, and annual energy estimation for the TFDL.
- An AutoCAD Single-Line Diagram documenting the electrical integration of the PV system with the existing 600V MSB.
- An ETAP power system model incorporating the TFDL electrical distribution network, grid connection, transformers, loads, and PV generators.

- Grid impact studies: load flow analysis, time-series simulation, reverse power flow, N-1 contingency analysis, and smart inverter (Volt-VAR and Volt-Watt) evaluation.
- An economic and environmental analysis including LCOE, NPV, IRR, and $CO_2$ reduction calculations.
- A low-voltage physical prototype for conceptual validation of voltage measurement, current sensing, power calculation, and simulated inverter response.

**7.2 Final Product Functionality**

The final deliverables achieve the following functionality:

- PV System Design (OpenSolar): The designed system with 661 × 550W Axitec modules and seven inverters generates 487,283 kWh annually, offsetting 27–37% of TFDL's annual grid energy demand. The system is fully string-sized and ready for procurement and installation planning.
- AutoCAD SLD: The completed SLD documents the full electrical integration from rooftop DC arrays through inverters, generation metering, isolation, and connection to the MSB, providing a construction-ready reference document.
- ETAP Model and Grid Studies: The ETAP model provides quantitative analysis of PV system impact on voltage profiles, power flows, transformer loading, and system behavior under contingency conditions. The model is validated and confirmed compliant with grid voltage standards (0.95–1.05 p.u.).
- Smart Inverter Evaluation: Volt-VAR and Volt-Watt control functions were successfully implemented and evaluated, confirming their effectiveness in maintaining voltage stability under high PV penetration.
- Economic Analysis: LCOE of $0.079/kWh confirms economic competitiveness vs. Alberta's grid rate of $0.12/kWh. IRR of approximately 4.25% over 25 years confirms financial viability.
- $CO_2$ Analysis: Annual emissions reduction of 228 metric tonnes $CO_2e$ confirmed, demonstrating meaningful environmental benefit.
- Physical Prototype: A functional prototype was assembled and demonstrated, measuring real-time voltage, current, and power under three simulated voltage conditions, providing proof-of-concept for the inverter sensing and response logic.

**7.3 Deviations from Original Scope**

The following minor deviations from the original scope occurred and are justified below:

- Prototype Scope: The prototype was intended to provide full conceptual validation of inverter response behavior. Due to component constraints, the prototype operates at low voltage (≤30V DC) and does not simulate grid-level voltages. Its purpose is conceptual demonstration only, which was clearly identified in the original scope and is consistent with the project objectives.
- BESS Not Implemented: Battery Energy Storage System (BESS) integration was identified as a recommended next step rather than a required deliverable. Reverse power flow mitigation through BESS is proposed for future work and is not part of the current project scope.

## 8. Technical Specifications of the Final Product

### 8.1 PV System Specifications

| Parameter | Specification |
|---|---|
| PV Module | Axitec AXIbiperfect GQ TS 550W Bifacial N-Type TOPCon |
| Module Efficiency | 22.33% |
| Number of Modules | 661 modules |
| Total DC Capacity | 363.55 kW DC |
| Total AC Capacity | 384 kW AC |
| Module Warranty | 30 years manufacturer and performance warranty |
| System Voltage (max) | 1500 VDC |
| Module Open Circuit Voltage (Voc) | 43.61 V |
| Module Maximum Power Voltage (Vmpp) | 36.42 V |
| Module Short Circuit Current (Isc) | 15.96 A |
| Module Maximum Power Current (Impp) | 15.10 A |

| Parameter | Specification |
|---|---|
| Annual Energy Generation | 487,283 kWh/year (OpenSolar estimate) |

### 8.2 Inverter Specifications

| Inverter Model | Quantity | Rated AC Power | MPPT Channels | Max Efficiency |
|---|---|---|---|---|
| Solis S6-GC60K-US | 5 | 60 kW each | 6 MPPT / 12 strings | 98.7% |
| Solis S6-GC40K-US | 1 | 40 kW | 5 MPPT / 10 strings | 98.7% |
| Growatt 30000TL3-S | 1 | 30 kW | 2 MPPT | ~98.5% |
| Solis S6-EH2P14K03-SV | 1 | 14 kW | 2 MPPT | ~97.5% |

### 8.3 Electrical Grid Integration Specifications

| Parameter | Value |
|---|---|
| Grid Connection Voltage | 600V (3-phase, 4-wire) |
| Transformer Rating | 2000 kVA, 13.2 kV / 347–600 V |
| PV Interconnection Point | Dedicated backfeed breaker at 600V MSB |
| Protection Devices | DC isolators, AC isolators, lockable 2-pole breaker, kWh generation meter |

| Parameter | Value |
|---|---|
| Bus Voltage (PV ON) | 0.99 p.u. (acceptable range: 0.95–1.05 p.u.) |
| Transformer Loading Reduction | 40–47% reduction at average load conditions |
| Grid Energy Offset | 27% (winter) to 37% (summer) of annual utility demand |
| Reverse Power Flow (Critical) | Observed at minimum load + 100% PV (mitigated by smart inverter controls) |
| N-1 Contingency | System stable with TX1 out of service; TX2 handles full load without overload |

**8.4 Physical Prototype Specifications**

| Component | Specification |
|---|---|
| Microcontroller | Arduino Nano ESP32 |
| Voltage Sensor | ZMPT101B (resistive divider-based measurement) |
| Current Sensor | ACS712 (5A range, Hall-effect sensor) |
| ADC Resolution | 12-bit (0–4095 levels, 3.3V reference) |
| Sampling Method | Averaging of multiple ADC samples per reading |
| Calibration | Automatic zero-offset calibration at startup |
| Noise Handling | Dead-band threshold applied for near-zero current |
| Power Calculation | Instantaneous power computed as $P = V \times I$ |
| Power Supply (Test) | Jesverty SPS-3010N (0–30V DC, 0–10A) |

| Component | Specification |
|---|---|
| Load Type | Resistive load (150Ω, 5W) |
| Output Interface | Serial monitor (115200 baud) |
| Visual Indicator | LED for power flow detection |
| Total Prototype Cost | Approximately $159.41 CAD |

## 9.Measuring Success and Validation Test Results

### 9.1 Success Criteria and Results

| Success Criterion | Target | Achieved | Status |
|---|---|---|---|
| PV System Capacity | 300–600 kW | 363.55 kW DC / 384 kW AC | Met |
| Annual PV Energy Output | 400–800 MWh/year | 487,283 kWh/year (487.3 MWh) | Met |
| Bus Voltage (PV ON) | 0.95–1.05 p.u. | ~0.99 p.u. (all scenarios) | Met |
| Voltage Rise | ≤ 5% | No violation observed in any scenario | Met |
| Transformer Loading | ≤ 90% | Reduced by 40–47% with PV (max 87.28% without PV) | Met |
| Grid Energy Offset | ≥ 10–15% | 27% (winter) / 37% (summer) | Exceeded |

| Success Criterion | Target | Achieved | Status |
|---|---|---|---|
| LCOE | < $0.12/kWh (grid rate) | $0.079/kWh | Exceeded |
| $CO_2$ Reduction | Quantitative estimate | 228 metric tonnes $CO_2e$/year | Met |
| N-1 Contingency Stability | System remains operable | TX2 handles full load; no voltage violations | Met |
| Smart Inverter Validation | Volt-VAR & Volt-Watt functional | Both implemented and validated in ETAP | Met |
| Prototype Demonstration | Conceptual validation | Functional prototype built and tested | Partially Met |

### 9.2 Validation Tests

#### 9.2.1 ETAP Load Flow Validation

Load flow validation was conducted by comparing simulated results against expected theoretical behavior based on the building's known load data and PV system sizing:

- With no PV generation, simulated utility supply of 463.1 kW closely matched the building's recorded average demand, confirming model accuracy.
- As PV penetration increased from 0% to 100%, utility demand decreased in a monotonically consistent manner, and transformer loading reduced proportionally, confirming correct power balance modeling.
- Bus voltages remained within 0.99–1.00 p.u. across all PV penetration levels, confirming grid voltage compliance.

#### 9.2.2 Time-Series Load Flow Validation

Time-series simulations for January 8 and June 8 were validated against the expected seasonal solar irradiance behavior:

- Summer simulations (June 8) showed higher PV contribution and greater utility demand reduction than winter simulations (January 8), consistent with Calgary's seasonal solar irradiance patterns.
- The 27% peak demand reduction (winter) and 37% reduction (summer) are consistent with the system's 363 kW capacity relative to TFDL's approximately 600 kVA peak demand.
- Negligible PV contribution at 6:00 AM and 6:30 PM was confirmed as consistent with sunrise/sunset times for Calgary in January and June.

### 9.2.3 Smart Inverter Validation

The Volt-VAR and Volt-Watt control functions were validated by comparing PCC voltage profiles before and after enabling smart inverter controls:

- Without smart inverter controls, simulated voltage rise under high PV penetration approached levels that could stress grid equipment.
- With Volt-VAR control enabled (user-defined Q(V) curve), reactive power injection/absorption effectively regulated bus voltage within ±1% of the nominal value.
- With Volt-Watt control enabled, active power curtailment was triggered when PCC voltage exceeded 1.03 p.u., preventing violations of the 1.05 p.u. grid limit.
- PCC voltage remained stable at approximately 0.99 p.u. in all scenarios with smart inverter controls enabled, confirming successful voltage regulation.

### 9.2.4 Physical Prototype Validation

The physical prototype was validated through a series of controlled experiments designed to evaluate its ability to measure voltage, current, and power under different operating conditions. The objective of this validation was not to replicate full-scale grid behavior, but to confirm the correct functioning of measurement and monitoring principles.

Three test scenarios were implemented using the adjustable DC power supply to simulate varying input conditions:

| Test Condition | Supply Voltage (V) | Measured Voltage (V) | Measured Current (A) | Measured Power (W) | Result |
|---|---|---|---|---|---|
| Low Voltage | 8.5 V | 8.5 V | 0.04–0.06 A | 0.30–0.50 W | Pass |

| Test Condition | Supply Voltage (V) | Measured Voltage (V) | Measured Current (A) | Measured Power (W) | Result |
|---|---|---|---|---|---|
| Normal Voltage | 12 V | 11–12 V | 0.06–0.08 A | 0.60–0.90 W | Pass |
| High Voltage | 15 V | 14–15 V | 0.08–0.10 A | 1.00–1.50 W | Pass |

Measured values closely followed expected theoretical relationships based on Ohm's Law, confirming correct sensor operation and data processing.

Noise in the current sensor was effectively mitigated through sample averaging and zero-offset calibration. The implementation of a dead-band threshold eliminated false readings when no current was present. Voltage readings remained stable across all test conditions, indicating reliable scaling and ADC performance.

The prototype successfully demonstrated real-time monitoring of electrical parameters and basic system response to varying input conditions. While it does not represent full-scale inverter operation, it provides a valid conceptual validation of the measurement and control principles used in grid-integrated PV systems.

**9.3 Economic Validation**

Economic feasibility was validated through the following calculations:

- Total Capital Cost (CAPEX): 363 kW × $2.0/W = $726,000 (rounded to $768,000 including installation balance of system costs).
- Annual O&M Cost: 1% of CAPEX = $7,680/year.
- Total 25-Year Cost: $768,000 + ($7,680 × 25) = $960,000.
- Total 25-Year Energy Production: 487,283 kWh × 25 = 12,182,075 kWh.
- LCOE: $960,000 / 12,182,075 kWh = $0.079/kWh - significantly below Alberta's commercial electricity rate of $0.12/kWh.
- IRR: Solved iteratively at approximately 4.25% - positive and above the 0% break-even threshold, confirming financial viability.

- Annual $CO_2$ Reduction: 487,283 kWh × 0.470 kg $CO_2e$/kWh = 228,923 kg = approximately 228 metric tonnes $CO_2e$/year.

## 10. List of Tools, Materials, Supplies, Costs, etc.

### 10.1 Software Tools

| Tool | Purpose | Cost |
|---|---|---|
| OpenSolar | PV system layout, string sizing, energy estimation | Free (academic) |
| ETAP (Student Edition) | Power system modeling, load flow, grid impact analysis | Free (student license) |
| AutoCAD (Student Edition) | Single-Line Diagram development | Free (student license) |
| Arduino IDE | Microcontroller programming for prototype | Free |
| Microsoft Office | Report writing, submission, and project coordination | Free |

### 10.2 Hardware Components - Physical Prototype

The physical prototype was assembled using commercially available, low-cost components selected to demonstrate key measurement and monitoring functions. A budget range of $80–210 CAD was initially estimated, and the final cost of $159.41 CAD falls well within this range.

| Component | Description | Estimated Cost (CAD) | Actual Cost (CAD) | Status |
|---|---|---|---|---|
| DC Power Supply | Jesverty SPS-3010N (0–30V, 0–10A) | $60–80 | $84.99 | Slightly above estimate |
| Arduino Nano ESP32 | Microcontroller unit | $20–30 | $33.43 | Slightly above estimate |
| Buck/Boost Converter | DC voltage adjustment module | $10–15 | $12.50 | Within estimate |
| Voltage Sensor | ZMPT101B module | $5–10 | $8.25 | Within estimate |
| Current Sensor | ACS712 (5A) | $5–10 | $7.50 | Within estimate |
| Resistive Load | 150Ω, 5W resistor | $10–20 | $9.99 | Below estimate |
| Breadboard & Wiring | Assembly components | $5–10 | $2.75 | Available |

The slightly higher cost of the power supply and microcontroller was offset by lower-than-expected costs in other components, resulting in no overall budget overrun. All components were readily available and required no specialized fabrication.

### 10.3 Full-Scale System Cost Estimate

| Cost Item | Basis | Estimated Cost |
|---|---|---|
| PV Modules (661 × 550W) | Market rate for commercial Axitec modules | $180,000 |
| Inverters (7 units) | Solis/Growatt commercial pricing | $120,000 |
| Installation and Balance of System | Electrical wiring, racking, conduit, labor | $300,000 |
| Engineering, Permitting, and Commissioning | Professional fees, CEC compliance | $120,000 |
| Monitoring and Metering | Generation meter, communication hardware | $20,000 |
| Contingency (10%) | Unforeseen costs | ~$74,000 |
| TOTAL CAPEX (estimated) | | ~$814,000 (~$2.0–2.25/W installed) |

## 11. Reflection on Ethics and Equity in your Capstone Journey

Throughout this capstone project, the team engaged with a range of ethical and equity considerations that shaped both the technical design decisions and the team's collaborative approach.

From an engineering ethics standpoint, the team was committed to accuracy, transparency, and professional integrity in all aspects of the project. This was particularly significant in the context of electrical power systems, where incorrect analysis or documentation could have real-world safety and financial consequences. To uphold these standards, all simulation inputs were verified against actual building data obtained from TFDL facility management, and all assumptions such as the lumped-load aggregation approach in ETAP - were explicitly documented and justified.

The team also reflected on the broader ethical implications of the technology being designed. Solar PV systems, while generally beneficial, involve supply chains that raise equity concerns: rare earth elements

and semiconductor materials used in photovoltaic modules are often sourced from regions with documented labor rights issues. The Axitec modules selected for this project are manufactured in China, and while the modules carry internationally recognized quality certifications (IEC 61730:2023, IEC 61215:2021), the team acknowledged that comprehensive supply chain ethics were beyond the project's immediate scope but represent an important consideration for real-world implementation.

Equity in access to renewable energy was another dimension the team considered. Large-scale commercial and institutional PV installations, such as the one designed for TFDL, are relatively straightforward to finance compared to residential solar in lower-income communities. The team recognized that the skills and frameworks developed in this project could be applied to advocate for more equitable solar deployment policies in Alberta, including community solar programs, net metering access, and incentive programs for underserved populations.

Within the team itself, equity manifested in workload distribution and decision-making. The team made deliberate efforts to ensure all members had meaningful contributions across technical and non-technical tasks. The rotating project management structure with different team members leading different phases helped prevent concentration of both responsibility and learning opportunity in any single individual.

Adherence to professional codes of conduct was maintained throughout the project. Team members followed the principles of the Engineers Canada Code of Ethics: holding public safety paramount, acting with competence and diligence, and being transparent about the limitations of the work. The explicit documentation of assumptions, uncertainties, and scope limitations in this report reflects that commitment.

**12. Reflection on Impact of Engineering on Society/Environment**

The proposed TFDL solar PV system has significant potential for both positive societal and environmental impact, and the team reflected carefully on both dimensions throughout the project.

On the environmental side, the most direct benefit is the reduction of greenhouse gas emissions. Alberta's electricity grid remains among the more carbon-intensive in Canada, with a grid emission factor of 0.470 kg $CO_2e$/kWh as of 2024. The proposed system, generating 487,283 kWh annually, would offset approximately 228 metric tonnes of $CO_2$ equivalent per year. Over the 25-year system lifetime, this represents approximately 5,700 tonnes of avoided $CO_2$ emissions the equivalent of removing approximately 1,240 gasoline-powered passenger vehicles from the road for a year.

Beyond carbon emissions, the team also considered other environmental impacts. The manufacturing of PV modules and inverters involves energy-intensive processes and the use of materials with their own environmental footprints. However, the energy payback period for modern bifacial N-Type TOPCon modules is typically 1–2 years, meaning the system generates its own embedded manufacturing energy many times over during its lifetime. At end-of-life, PV module recycling infrastructure in Canada remains underdeveloped, and this represents a genuine environmental concern that the team noted should be addressed in future project phases.

The environmental impact of the installation process itself including rooftop work, conduit installation, and cable routing was considered minimal given the building's existing electrical infrastructure and the non-invasive nature of the rooftop mounting system.

From a societal perspective, the project contributes to the University of Calgary's institutional sustainability commitments and serves as a visible demonstration of large-scale renewable energy integration for other commercial and institutional facility operators in Alberta. Universities have a unique role in shaping public understanding of energy transitions, and a successful PV installation on the TFDL campus could influence energy decisions across the institution and beyond.

The economic benefits reducing TFDL's annual electricity bill, generating long-term cost savings, and improving energy security also have societal value by freeing institutional resources for core educational and research activities. The LCOE of $0.079/kWh represents a meaningful cost advantage over the current grid rate of $0.12/kWh.

One negative societal consideration the team identified is the risk of grid destabilization if large-scale distributed PV adoption occurs without adequate grid planning and infrastructure upgrades. The team's ETAP analysis directly addressed this concern by demonstrating that with smart inverter controls (Volt-VAR and Volt-Watt), the proposed system operates safely within voltage and power flow limits. The team recommends that any large-scale commercial PV deployment in Alberta be accompanied by grid impact studies of the type performed in this project.

**13. Reflection on Team's Approach to Project Management**

The team's approach to project management evolved significantly across the three phases of the project Design Roadmap, Progress Reports, and Final Report and the team reflects on both its strengths and areas for improvement.

At the outset, the team established a clear work breakdown structure aligned with the project deliverables: site assessment, PV design, electrical integration, ETAP modeling, prototype development, economic analysis, and documentation. Responsibilities were assigned based on individual strengths and interests, with Pavithiran Babu focusing on ETAP modeling and power system analysis, Syed Muhammad Zulqarnain on PV system design in OpenSolar, Leo Osea on AutoCAD SLD development, Tahmina Tabassum Tonni on documentation and coordination, and Shafkat Haque on string sizing validation and technical review.

One of the most effective project management decisions was the establishment of fixed weekly team meetings. Early in the project, scheduling conflicts among team members due to overlapping course commitments and part-time work schedules created coordination challenges. The introduction of a standing weekly meeting time, combined with a shared Google Drive for document collaboration, significantly improved communication and progress tracking.

Overall, the team's project management approach was effective. The combination of structured task allocation, clear communication channels, risk tracking, and regular progress reviews enabled the team to deliver a comprehensive technical project within the allotted timeline and budget.

**14. Reflection on Team's Approach to Economics and Cost Control**

Economic considerations were integrated into the project from the earliest stages, beginning with the prototype component budget defined in the Design Roadmap and extending through the final LCOE and IRR calculations for the full-scale system.

For the physical prototype, the team established an estimated budget range of $80–210 CAD based on market research for the required components. Actual procurement costs totaled $159.41 CAD — within budget. Two components (the DC power supply at $84.99 and the Arduino Nano ESP32 at $33.43) slightly exceeded their individual upper estimates, but these overruns were offset by savings on the load module and the availability of breadboard materials at no additional cost. The team's decision to source components externally and track costs against the roadmap estimates reflects sound cost control discipline.

For the full-scale PV system, the team adopted a cost estimation methodology consistent with Canadian commercial PV industry benchmarks. The assumed installed cost of $2.0/W is consistent with utility-scale and commercial PV pricing in Western Canada as of 2024–2025. The team was transparent about the

uncertainty in this estimate actual costs for a real TFDL installation would depend on detailed contractor quotes, structural assessments, and permitting costs, all of which were outside the project scope.

The LCOE calculation of $0.079/kWh used a 6% discount rate, which the team selected as a representative cost of capital for institutional projects of this type. Sensitivity to the discount rate was implicitly acknowledged a higher discount rate would reduce the NPV and increase the effective LCOE. The team recommends that any real-world implementation of this project include a full financial sensitivity analysis varying the discount rate, electricity rate escalation assumptions, and O&M cost projections.

The IRR of approximately 4.25% was calculated iteratively and reflects a modest but positive return on investment appropriate for an institutional renewable energy project where risk tolerance is relatively low and where non-financial benefits (carbon reduction, energy security, institutional reputation) also contribute to the investment case. The team noted that federal and provincial incentives for commercial solar, including the Canada Greener Buildings Fund and potential carbon pricing benefits, were not fully modeled in this analysis and could improve the economic case significantly.

In terms of cost control culture, the team maintained consistent cost awareness throughout the project. No unplanned expenditures were incurred. All software tools used were either free academic licenses or university provided. This approach kept total project costs minimal and appropriately focused resources on analysis and documentation quality.

**15. Reflections on Team Member's Approach to Life-long Learning**

The capstone project provided each team member with substantial opportunities for learning beyond the formal curriculum in both technical depth and professional skills and the team reflects on how these experiences will contribute to ongoing professional development.

**Pavithiran Babu - Power System Modeling and ETAP**

Pavithiran entered the project with foundational knowledge of power systems but had limited hands-on experience with ETAP simulation software. The process of building the TFDL ETAP model from scratch developing the grid topology, configuring transformer parameters, setting up PV generators, and running load flow, time-series, and contingency studies represented a steep but highly rewarding learning curve. Pavithiran plans to pursue further training in ETAP and power system protection as part of his professional development in the electrical utility sector.

**Tahmina Tabassum Tonni - Technical Communication and Project Coordination**

Tahmina identified early in the project that her primary contribution would be in documentation quality, report structure, and team coordination. Through the process of writing and editing technical reports for a specialized engineering audience, she significantly developed her technical writing skills learning to translate complex simulation results and engineering analysis into clear, precise, and well-organized written communication. She also developed skills in project scheduling and stakeholder communication through her coordination with the TFDL facility management team. These skills directly apply to her career goals in engineering management and technical consulting.

**Shafkat Haque - String Sizing, MPPT Optimization, and Technical Review**

Shafkat's primary learning focus was the detailed technical process of PV string sizing and MPPT optimization. Working through the DC/AC ratio constraints for seven different inverters, each with different MPPT channel configurations and string voltage requirements, required developing a systematic analytical approach. He also served as the team's technical reviewer, developing skills in critically evaluating engineering work against industry standards. Shafkat identified a knowledge gap in his understanding of grid protection coordination, which he plans to address through independent study and professional development courses in his first years of professional practice.

**Leo Osea - AutoCAD and Electrical Drawing Standards**

Leo's primary contribution was the development of the AutoCAD Single-Line Diagram, an area he had studied formally but had limited real-world project experience with. Working from the TFDL E600 record drawing a professional-grade construction document and developing the PV integration SLD to the same standard required learning AutoCAD's electrical symbol libraries, CEC-compliant drawing conventions, and layering standards for electrical construction documents. Leo identified that understanding the relationship between design documents and actual electrical installations is a skill that requires ongoing development through field exposure, which he plans to pursue through co-op and internship opportunities.

**Syed Muhammad Zulqarnain - PV System Design and OpenSolar**

Syed led the PV system design in OpenSolar, which involved learning the platform's module library, inverter configuration tools, shading analysis features, and energy simulation engine. A key learning for Syed was the sensitivity of energy generation estimates to input parameters small changes in panel tilt, azimuth, or string configuration could produce meaningfully different annual output projections. This

experience reinforced the importance of uncertainty quantification and sensitivity analysis in engineering feasibility studies. Syed plans to pursue formal training in renewable energy system design and financial modeling to complement his technical skills.

**Team-Wide Learning**

As a team, the project reinforced several cross-cutting lessons: the value of early stakeholder engagement (the connection with TFDL facility management proved essential for obtaining accurate load data), the importance of iterative validation (each component of the project was checked against expected behavior before being used as input to the next analysis stage), and the necessity of clear written documentation not just as a final deliverable but as an ongoing tool for maintaining shared understanding within the team. These lessons will remain relevant throughout each member's professional career in electrical engineering.

**Appendices:**

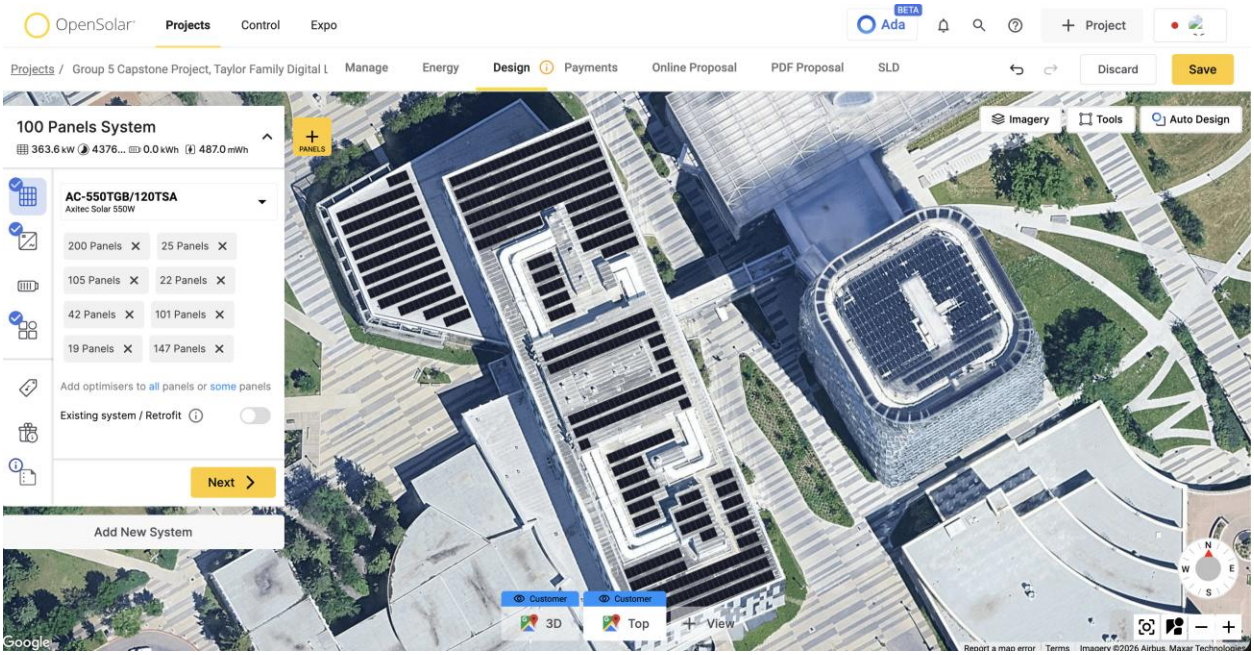



Figure 1: *PV modelling in Open solar*

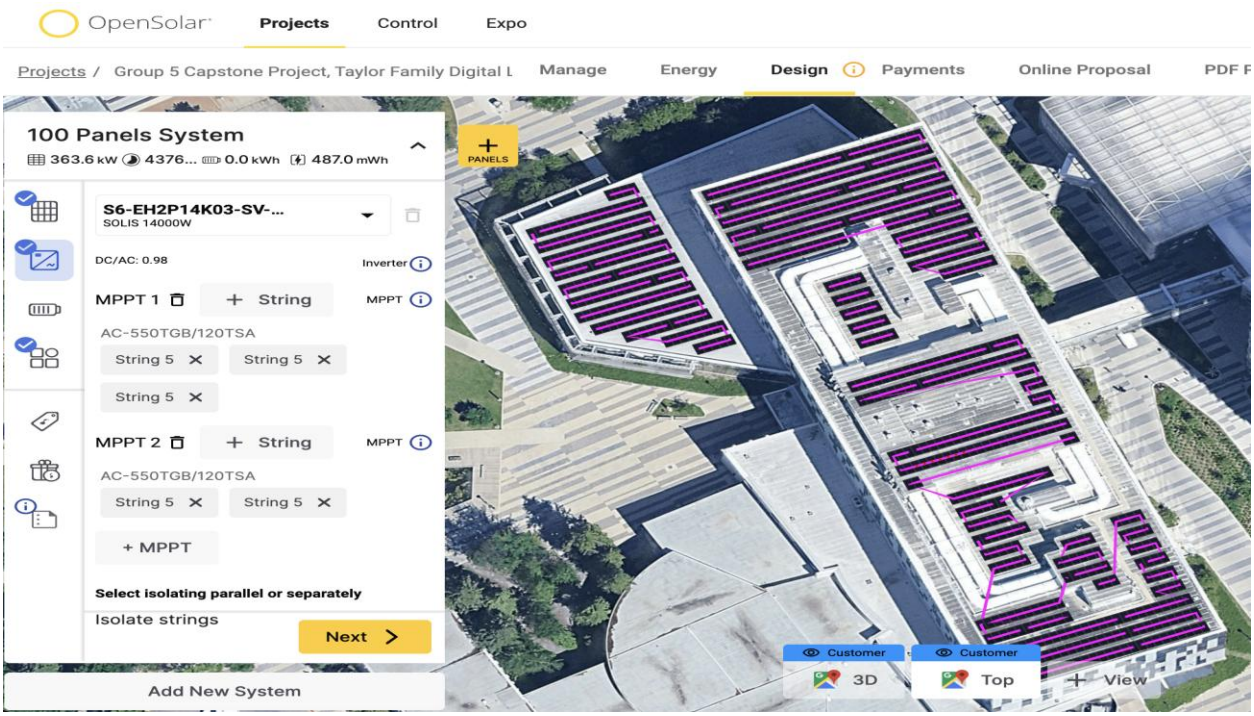



Figure 2: *Inverter modelling in Open solar*

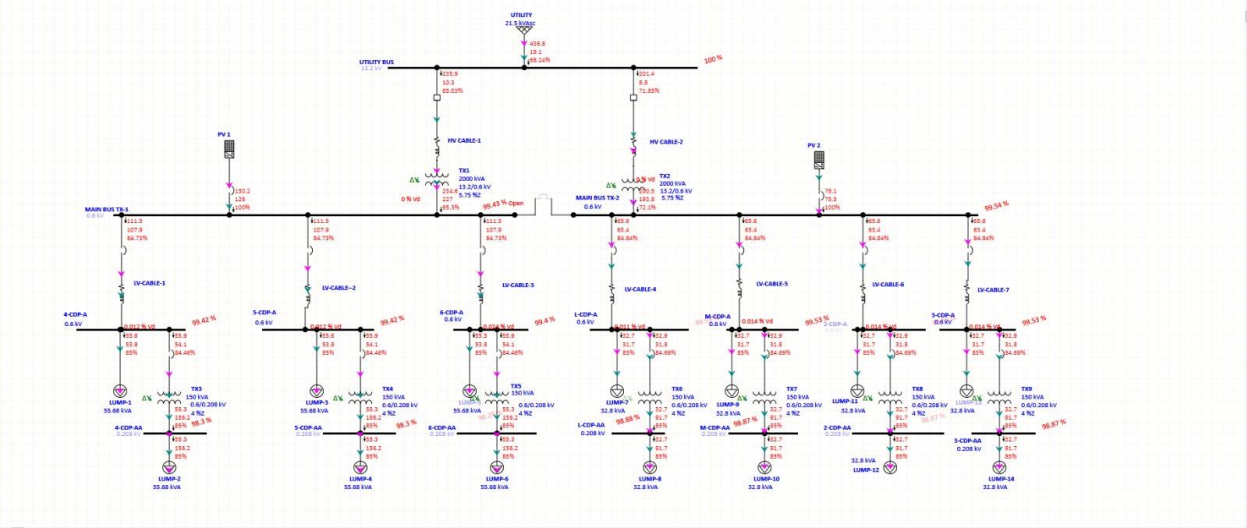



Figure 3: *ETAP Model with PV Connected*

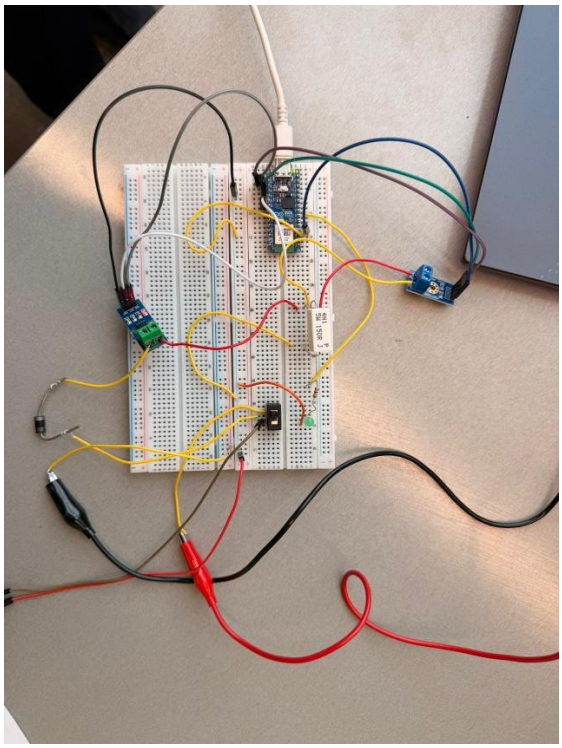


Figure 4: *Small Prototype*

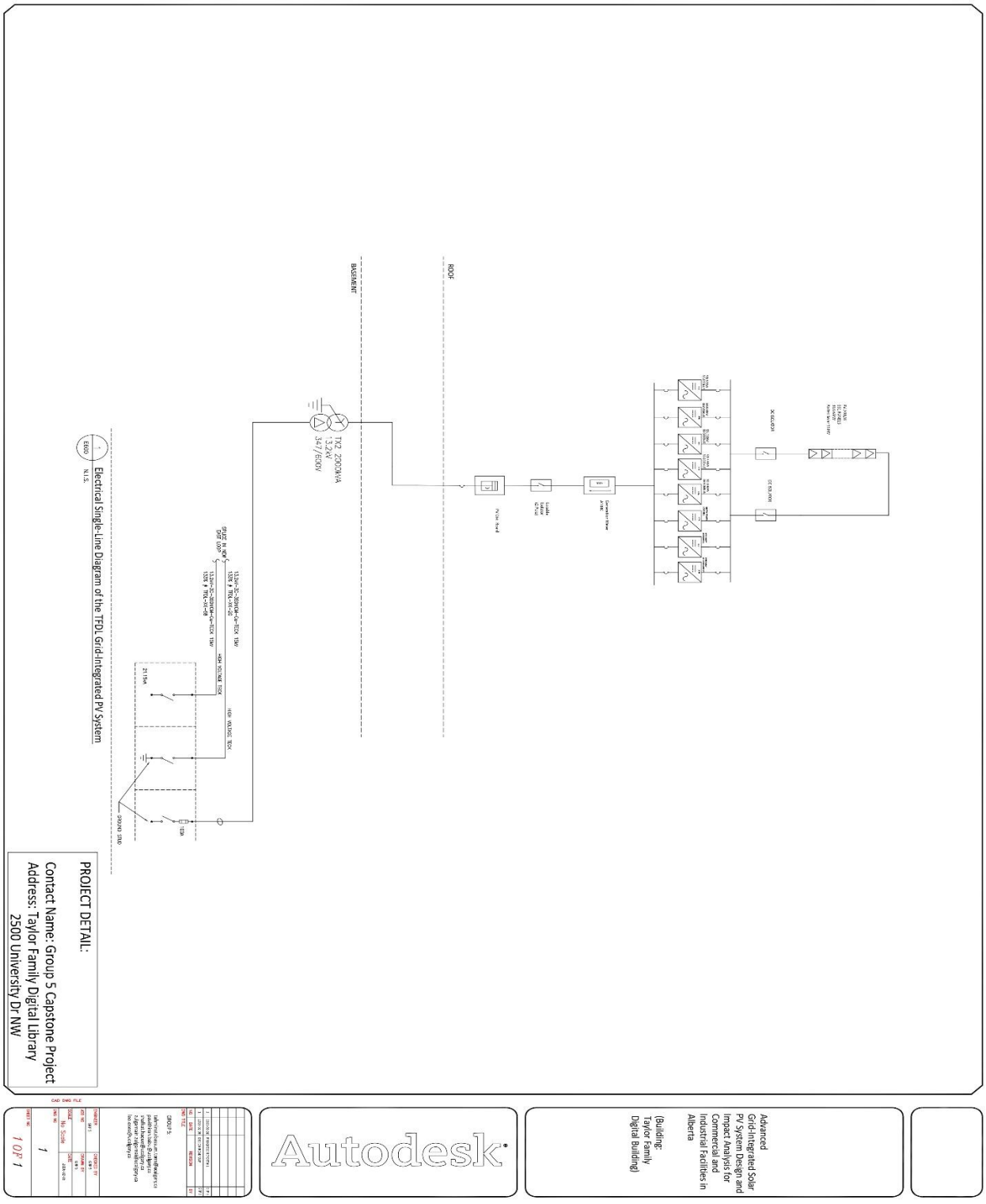



Figure 5: *PV connection at PCC*

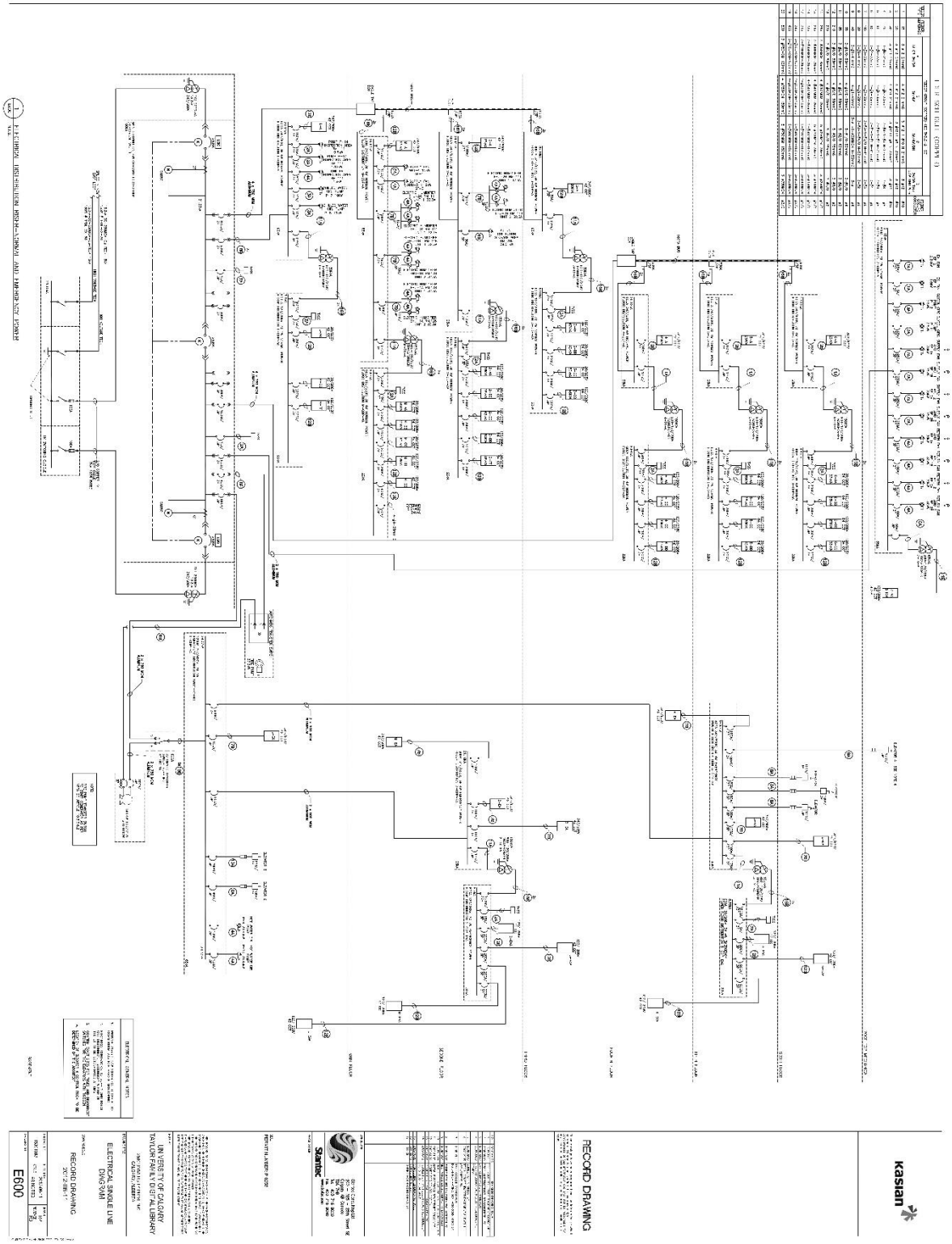



Figure 6: *TFDL SLD*

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